

Improving customer service through speech technology

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Interactive voice response (IVR) technology is playing a major role in the emerging call centre market-place, by linking telephone calls directly into company computer systems and enabling automatic processing of transactions. This paper describes the automatic customer interface (ACE) system, which is being developed for BT's personal communications division in support of its customer service operation.

1. Introduction

The rapid growth in telesales, telemarketing, help-desks, and services such as telephone banking, has raised the importance of customer call handling for many companies. In support, technology is providing more and more sophisticated tools, for both agents and management, in order to improve productivity, the effectiveness of customer contacts, and the quality of the customer experience.

Call centres [1, 2] are increasingly being used by major companies to focus telephone-based customer contact, and aid corporate process re-engineering. The aim is to make systematic use of the telephone in order to achieve key business objectives.

At their most effective, call centres can be thought of as the key points within the company at which telephone contacts are undertaken and managed. BT operates one of the largest call-centre operations in Europe, managing a wide variety of incoming customer calls for services such as operator and directory (national and international) assistance for over 23 million customers, customer service, as well as inbound and outbound telemarketing and telesales services.

Within a call centre, maximising the productivity of the agents is seen as a key goal. Agents are important to a successful operation — building relationships, growing the business and handling customer problems sympathetically, for example. People-related factors, however, generally dominate system and networking costs. As many companies strive to give better and better service, it becomes vital that skilled human resource is focused on the high-value activities. Technology is playing an increasingly important role in call-centre operations by taking over more of the routine aspects of processes.

Interactive voice response (IVR) systems provide a mechanism by which customer contacts can be fulfilled in either a fully or partially automated manner. By prompting a customer to enter information using either TouchTone[®] signalling (via a telephone or keypad) or speech input, an IVR system can carry out both routing to a required service, and completion of transactions automatically. This is particularly useful for simple interactions with callers where the involvement of human agents offers limited additional value, and out-of-hours services where agents may not be available.

Development of an IVR platform and services to support BT's customer service function has been under way for some time. The automatic customer interface (ACE) platform is based upon the interactive speech application platform (ISAP) [3], and has been operational since 1994 supporting trial and pilot services.

The processes supported by the system involve interaction with BT's customer service system (CSS) databases, and include the automatic placement of orders. The integration of new processes and working practices with existing large corporate 'legacy' data systems is typical of many call-centre IVR applications, and is a major area of implementation effort.

This paper discusses the integration of an IVR platform into call-centre and corporate data environments, and takes the ACE system as a case study. The development of two typical ACE services, an ordering service for BT products and services and a bill payment validation facility are discussed in detail. Objective and subjective results are presented together with ongoing developments.

2. IVR automation in call centres

In allowing calls to be routed efficiently to and between skilled agents with access to corporate data resources, call centres cut across the traditionally separate corporate worlds of voice telephony and computing. In the implementation of effective call centre operations, the distinct but related technologies of IVR [4] and computer telephony integration (CTI) [5] can be valuable tools.

IVR is itself a limited form of CTI, linking voice telephony and computing networks to enable calls to be handled automatically. A separate CTI controller within a call centre, however, also allows calls to be managed and routed intelligently between agents and IVR systems. The voice call, with related call and customer information, can be handled in a co-ordinated fashion throughout a complex call flow potentially involving more than one agent.

Figure 1 shows the key components which comprise an IVR enabled call centre and how CTI serves a key role in linking voice and data functions.

2.1 Call delivery and distribution

Calls are typically delivered to a call centre by the network provider in conjunction with number translation and billing services, such as those provided by BT telemarketing services. This allows companies to take

advantage of call stimulation and enhanced tariffing options such as Freephone or Lo-call (0800,0345, etc).

An automated call distributor (ACD) is employed to distribute calls between and among groups of agents in order to balance agent loading, caller wait time and in the case of 'blended call centres', the mix of inbound and outbound calls. For example, queues can be established to segment agents into different agent skill groups or campaigns and used to correctly route calls based upon factors such as call origin or the original number dialled. Typically, the ACD will also provide facilities such as music on-hold and position or time remaining in queue information.

2.2 Interactive voice response (IVR)

IVR can be used to automate fulfilment of selected queues, to provide call steering (for example, between departments) or to provide overflow capacity during peak periods. A proportion of customer needs can therefore be fulfilled with a minimum of human intervention allowing increasing flexibility and cost-effectiveness to be introduced into the call processing.

Typically customer contacts handled by telephone agents can be conducted at significantly lower cost than those by other manual means, as information can be entered directly by the agent and many transactions completed and

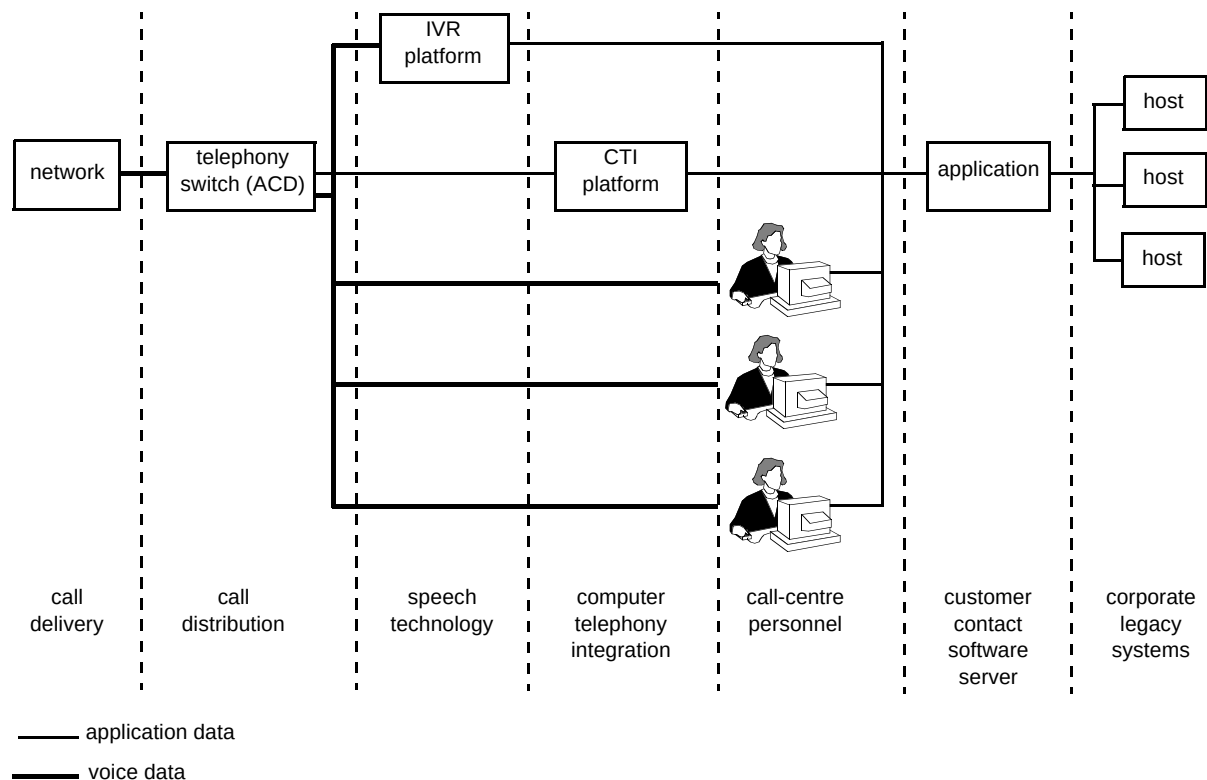


Fig 1 IVR and CTI within call centres.

fully documented immediately. Similarly, contacts fulfilled through IVR can be a further order of magnitude more cost effective.

In addition, IVR provides a mechanism for handling peaks in call rate effectively. For example, in many financial applications, call densities can be dramatically concentrated in relatively short peaks at predictable times through the day. If many of these can be handled through IVR, the existing call centre agents can manage traffic variations without the need for an additional tidal workforce.

Developments in speech recognition technology will continue to allow progressively complex applications to be automated. Typically, the benefits achieved through the appropriate use of IVR include:

- skilled agents can be focused on high value activities,
- agents can be moved from reactive in-bound call handling to proactive out-bound,
- 24 hour, 7 days per week working,
- ability to handle traffic peaks,
- flexibility to target new market opportunities,
- enables new commercial channel for services which would not otherwise be cost-effective,
- deliver services at lower cost.

As with other technologies, IVR should be regarded as a business tool, and care should be taken in choice of applications. Inappropriate use may result in negative customer reaction, which, while reducing costs in the short term, may damage long-term relationships. IVR technology and the effective design of both TouchTone and speech recognition-based customer dialogues and systems are discussed in more detail in Whittaker and Attwater [4].

2.3 Computer telephony integration (CTI)

CTI [5] allows the co-ordinated control and manipulation of call (voice) and call or customer related data in order to provide more flexible and effective call flows. For example, within a CTI application, a call might be passed between agents (e.g. a customer service agent and a specialist, or a sales agent and an overdue account agent) along with information such as the identity of the caller. This allows both agents access to that caller's records within the company's corporate data, and information about the transaction to date.

CTI allows the use of telephony-related information such as calling line identity in conjunction with application

or corporate data, such as customer records — for example, to allow the agent's screen to be pre-populated with the relevant call information. This process of 'screen popping' ensures that, when agents either receive new calls or are passed one by another agent, they have access to all of the call and customer-related information available in order to handle the call and fulfil the customer's requirements.

2.4 Network-based call centres

Traditionally, call centres and IVR functions have been implemented as bespoke solutions situated in customer premises, in which calls are routed to the centre by the network operator who provides facilities such as number translation, flexible billing and time-of-day routing. However, it is now possible to provide increasing levels of managed-service support within the network. This offers an evolutionary path from existing business practices and can provide a number of advantages:

- reduced capital and manpower investment,
- guaranteed upgrade path as new generations of technology are deployed,
- advanced network management and resilience,
- capability to deal with tidal and changing traffic patterns.

3. ACE platform and service development

A significant number of BT's business transactions with its residential customers are conducted over the telephone by customer service advisers based in a number of regional call centres. These transactions include telephone account management, product ordering and fault reporting. A significant proportion of these calls are low-complexity transactions. The ACE platform is aimed at automating calls of low complexity, enabling customer service advisers ('agents') to concentrate on work of a higher value, both for customers and BT.

Many of the activities carried out by the customer service advisers involve interaction with BT's customer service system (CSS) databases. CSS maintains information on every BT telephone installation in the UK, and incorporates processes for transactions such as the ordering of new network products, and the retrieval of customer installation details. The ACE system integrates with CSS to perform many of these functions without human intervention.

The following section describes the ACE system installed in London on which services are trialled and piloted. A national platform is currently being installed, which will allow services which have been piloted

successfully to be installed and fully supported by BT's operational support units.

3.1 Platform infrastructure

Figure 2 details the architecture of the London installation. Calls are routed over the BT derived-services network to a Nortel DMS100 switch which forms a component of BT's customer-handling voice network. They are then switched to the ISAP over a digital PCM trunk using a digital private network signalling system (DPNSS) interface. As the ACE applications do not require the full functionality of the CCITT No. 7 signalling system, the use of DPNSS allows systems to be launched with less stringent testing requirements and, as a consequence, significantly reduced development time-scales and costs.

The ISAP provides the required IVR functionality such as TouchTone recognition, message play-out and dialogue management. Where appropriate, calls can be routed to agents via the DMS100. The BT VISAGE service creation tool [6] is used to develop the IVR dialogues. Although the ACE applications described here are implemented using TouchTone-based dialogues, the advanced speech technology supported by the ISAP and VISAGE provides a path for the progressive automation of increasingly complex services within the same architecture.

The IVR interacts with the CSS database through a software interface referred to as the customer host machine interface (CHMI), which abstracts detailed host interactions from individual application implementations. Once proven, the automated business processes can be migrated to corporate process interfaces such as the customer handling intermediate server (CHIS). For resilience, data is routed to two remote CSS sites over separate X.21 circuits. If access is required to accounts outside of the host database area, transaction switching is used to route the data appropriately.

A management information system allows the collection of information relating to the services. This supports both the collation of regular sales and marketing reports on orders raised through ACE, and the generation of objective measures used to assess the effectiveness of the services and customer dialogue.

A management console interface and visual alarm indicator are provided together with a radio-pager interface for use in alarm conditions.

3.2 ACE service design

The design of ACE services is based on the guidelines of BT's internal voice application style guide and follows a rapid application development approach. This approach relies on the intensive assessment and iterative improvement of a series of prototype systems. As many of

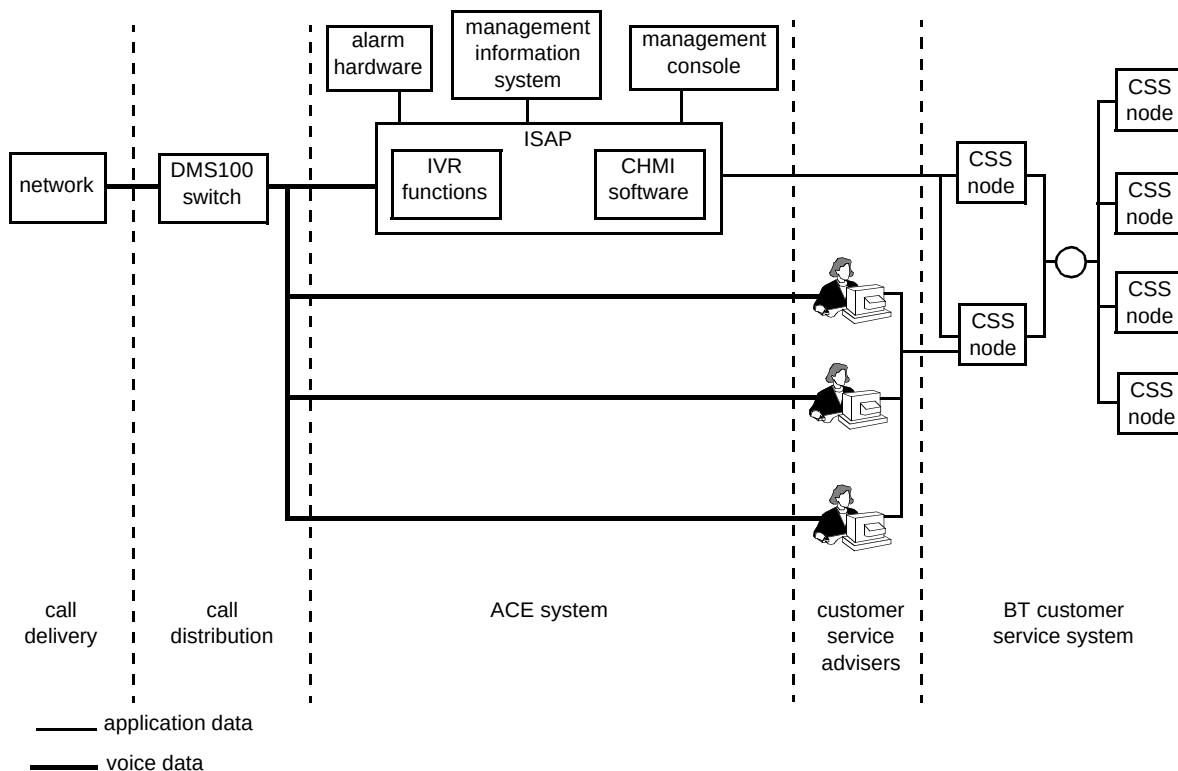


Fig 2 ACE platform architecture.

the ACE services involve fairly short, straightforward dialogues and receive high call rates, the technique can be employed effectively.

At an early stage of development, 'concept testing' of ACE services was carried out with potential users. This highlighted a number of factors which were carried forward into the dialogue designs for the individual applications. For example, a general awareness and acceptance of speech-based services was noted.

Initial user requirements included:

- clear and easy-to-use instructions,
- simple procedures to follow,
- ease of use,
- fallback to a human operator if needed — even for relatively straightforward service interactions.

Requirements capture for each of the service concepts identifies issues such as segmentation of potential users, type of tasks to be carried out, frequency of use, and standard terminology to be used. Data from this stage is used to drive the design of the front-end dialogue for the service, as well as, for example, the underlying procedures for interaction with databases and correction-strategy requirements. Initial requirements for priming material and customer service adviser handling procedures can be determined at this time.

Within the UK, the usage of TouchTone phones by residential callers to BT is around 50%—60%. Services aimed at the residential market must take non-TouchTone callers into account in service design.

3.3 ACE services

This section describes two typical services which have been piloted on the London installation. The first service, 'Paid-My-Bill', has been designed for use by residential customers, and the second service 'Third-Party Retailer' has been designed for a small, closed-user group.

'Paid-My-Bill' service

There may be a number of reasons why payment of a residential bill may not have been received by the due date — for example, the payment may have been lost in the post, delays in processing of the payment, or the customer has delayed payment. If payment is not received within a certain period after issue of the bill, a reminder letter is issued. This includes a telephone number for account enquiries.

A high proportion of calls resulting from this letter are from callers who would like to inform BT that they have

paid or are about to pay their bill. These are generally straightforward calls which are suitable for automation.

For the pilot service, a distinct telephone number was given to distinguish calls stimulated in this way from more general billing enquiries. One of the goals of the dialogue is to identify those calls suitable for automation and to route other callers effectively and acceptably to customer service advisers for fulfilment. In general, as callers have received little or no priming on use of the system, and will be naive or infrequent users, the system dialogue is designed to guide them effectively through the call.

In order to filter those callers using TouchTone phones into the automated system, the service initially asks the caller to indicate that they have a TouchTone dialling telephone, by pressing the '*' button twice. This sequence has the effect of enabling the tone-dialling capabilities for certain types of dual-signalling telephone.

The caller is then prompted to select whether they have paid their bill, are about to pay their bill or if they wish to speak directly to a customer service adviser. If they elect the latter option, they are transferred to the appropriate regional customer service centre, which is identified from the access number initially dialled to access the service. For the other callers, the system checks the status of the customer's account on CSS. If the account is clear, the customer is informed, otherwise the details are processed and the account handled in line with normal business processes. Figure 3 provides a typical customer dialogue.

'Third-Party Retailer' service

This service supports the sale of BT network products and services by third-party retailers, for example high street retail outlets. It is aimed at a closed user group of staff in those retail outlets who receive training in the use of the service and who will typically access it on a regular basis. As a result, the dialogue can be more complex than that used for the 'Paid-My-Bill' service. Each retail outlet is assigned a retailer identification number, and regular reports are generated which show ordering details against each retail outlet. This proved to be a very effective way by which business processes in support of a new route to market could be rapidly and flexibly implemented and rolled out.

The dialogue prompts the caller to enter, using TouchTone, the retailer identification number followed by the telephone number of the customer. It then moves to a menu allowing the selection of a number of products for that customer. The dialogue loops to allow the retailer to order products for a number of customers within a single call. Figure 4 shows the dialogue flow for the service.

Good morning, welcome to BT's payment service.

If you have a tone dialling phone, I can deal with your call quickly. Otherwise, please wait for an adviser. If you have a star button, please press it twice, now.

<caller enters '**'>

Thank you. You now have three choices. If you are calling to say you have already paid your bill please press '1'; if you are about to pay your bill, please press '2'; to speak to an adviser, please press '3'.

<caller enters '1'>

Again, using the buttons on your phone, please key in the full telephone number including the dialling code.

<caller enters a telephone number>

I recognised the phone number as <number>. Using '1' for yes, and '2' for no, is this the correct number ?

<caller enters '1'>

Please wait a moment while I look up your account details. Please wait a moment longer. You are not being charged for this call.

I have checked your account, and it is clear. I have entered your call on our records.

Thank you for calling BT. If you would like to know more about our bill payment options and discount schemes, please call us free on 150.

Goodbye.

Fig 3 Example 'Paid-My-Bill' interaction.

3.4 Call traffic levels

During the analysis phase of the pilots, both ACE services described above processed over 5000 calls per week, including a high proportion of multiple orders for the ordering service.

4. Analysis of system performance

4.1 Assessment methodology

The methodology used to access services was based upon experience gained in the deployment of services such as CallMinder [7], and was developed, tested and proved during development of an early ACE service. In line with the iterative approach to system development, the process involved both the comparison of performance between successive iterations and the development of an overall picture for each service. For example, rewording of the initial prompt within the 'Paid-My-Bill' system resulted in a reduction of 50% in the rate of hang-ups.

The assessment focused on both subjective and objective measures. Subjective data was collated from help desk information, and questionnaires used during follow-up telephone interviews carried out by a market research agency. Subjective analysis investigated:

- 'usability' — the effectiveness and efficiency with which customers can use a service,
- 'acceptability' — whether a service meets customer expectations and the level of satisfaction resulting from using the service.

Objective usage data was collected through logging within the applications — for example, the distribution of interaction times with the service, the number of calls made to the service platform, the number of successfully automated calls, and the number of transfers to the help desk were logged.

4.2 'Paid-My-Bill'

Different aspects of 'Paid-My-Bill' were assessed as the service was modified over a number of iterations. Trials of an early version of the service highlighted a number of positive results. For example, subjective data showed that

83% of customers interviewed were very satisfied with the service, and 92% reported that the service was easy to use.

The option to transfer to an adviser was successful in effectively handling customers who had called other than to notify BT of bill payment, e.g. for complex account enquiries. Subjective analysis also showed that the attitude of callers differed very little whether their call was completed automatically or by a customer service adviser. Customers perceived the option to transfer to an adviser as a successful interaction with the ACE platform.

The 'Paid-My-Bill' system successfully filtered a high proportion of TouchTone users for automation. As an example, Fig 5 shows the performance of the application during a one week period of the trial — 95% of TouchTone calls were successfully processed with 22% of transactions fully automated.

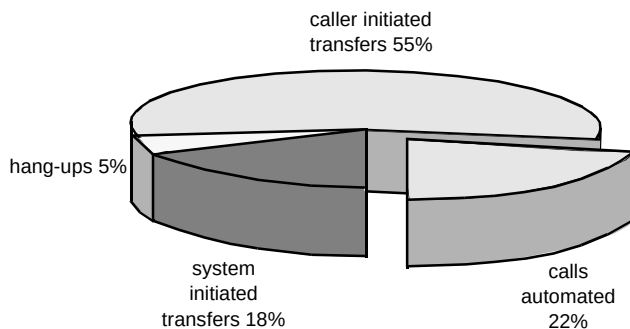


Fig 5 'Paid-My-Bill' — objective analysis.

4.3 'Third-Party Retailer' service

During development of the 'Third-Party Retailer' service, a simplified version was used to assess usability issues. This service allowed retailers to order only BT's 'Caller Display' service for customers, whereas the final system caters for a wider range of products.

Figure 6 details the subjective profile obtained from analysis of interviews with retailers.

Subjective data showed that, overall, attitude among the user group towards using automated services, such as the ordering line, is very positive. For example, 76% responded either very satisfied or quite satisfied with the system and 79% were confident that the system worked either very well or fairly well.

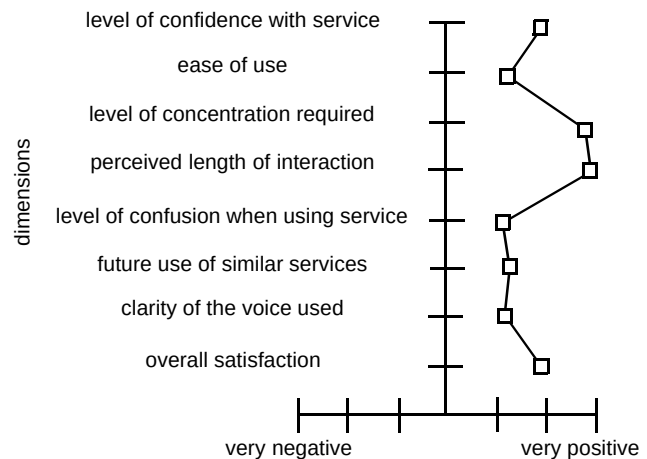


Fig 6 'Third-Party Retailer' — subjective analysis.

In response to the question: 'What did you particularly like about ACE?', the most frequent replies were that it was:

- easy to use,
- quick to use,
- efficient.

A number of usability issues were identified during the trial which were addressed in later iterations of the dialogue. For example, the trial was conducted shortly after the UK national code change. Relevant prompts were reworded to be more explicit after it was observed that some retailers were entering superseded area codes. Similarly the need to prime retailers to have the complete customer details to hand before calling the system became clear.

Subjective data highlighted that during busy periods retailers were using the service with high levels of background noise. As a result, the recording level of the system prompts was increased. Retailers also requested additional confirmation that the order had been placed

The findings of the caller display ordering trial were incorporated into the 'third-party retailer' dialogue, with only minor changes to the messages which had been optimised in the iterative development.

Objective data collected during a typical week of the trial showed that 97% of calls were handled without any operator fallback. Over 80% of orders placed were fulfilled automatically, with the remainder requiring manual intervention due to particular factors relating to the customer's installation or account.

4.4 *Lessons learnt from ACE trials*

The trials process within the ACE development has provided a large amount of valuable usage information and identified additional domain-dependent and independent usability issues. These have been fed in both to the further ACE services and non-BT call centre applications.

Particular observations can be made about the differences between services aimed at members of the public who may have little experience of speech systems, and closed user groups who can be educated through priming material and experience. The requirements for these groups may be significantly different and must be taken into account during dialogue design.

As discussed in section 3, one of the factors limiting the proportion of calls which a TouchTone dialogue can automate for residential applications, is the incomplete penetration of TouchTone within that market segment. Systems based upon speech recognition offer significant benefits in this area and can be offered using the functionality of the ISAP already employed within the ACE architecture. Speech recognition and mixed modality dialogues have been designed and employed within the ACE application area but are not discussed here.

5. **ACE future development**

The success of the trials described above underlines the effectiveness of IVR technology within a call centre environment. Through careful service design, applications can address the needs of both the general public and also specialised closed user groups. The range of services that can be targeted is growing as both TouchTone penetration increases and speech recognition technology improves. In addition, the rapid growth in the Internet is likely to mean that on-line customer service will take on increasing importance in future.

The potential for use of ACE services within BT's customer service operation is large, both for automating existing business processes and as a cost-effective mechanism for introducing new channels to market. An ongoing programme of development is under-way, with a number of new services under evaluation.

A national platform is currently being installed to support ACE services. This platform integrates the ACE service functionality into BT's corporate voice and data network infrastructure, and operational support processes.

6. **Trends in IVR call-centre technology**

Call centres are becoming increasingly important as one of the primary ways in which companies interact with their customers, and this paper has highlighted how IVR technology can be integrated to automate simple transactions and bring business benefit.

As customer handling becomes a major corporate issue, many companies are seeking to integrate their geographically and organisationally disperse call centres to work more closely and act as a large, virtual unit. Call centre technology is also being applied internally to enable flexible working via applications such as hot desking and remote working.

Within a call centre today, voice and data are typically handled separately. The integration of high bandwidth technologies such as ATM (asynchronous transfer mode) into the infrastructure will allow voice, data, images and video to be handled in a seamless fashion. This will occur in parallel, with increasing variety in the ways in which customers will want to interact — on-line through systems such as the World Wide Web, via desktop videotelephony or through video and data kiosks in airports, hotels and retail outlets.

This paper has concentrated on the use of TouchTone-based IVR applications. Developments in speech technology will continue to shift the emphasis of many applications towards dialogues where voice is the primary or sole mode. Speech recognition is of particular relevance both in situations where TouchTone penetration is low and where the complexity or inherent vocabulary do not map well on to a TouchTone dialogue, e.g. the entry of names.

The availability of rapid vocabulary creation using sub-word-based recognition enables designers to build recognisers with vocabularies suited to the particular application, such as product names, allowing a move away from menu-based dialogues into more flexible structures which will open new applications to the possibility of automation.

Developments in robust large vocabulary speech recognition and in the quality of text-to-speech systems will enable a new generation of advanced IVR solutions, for example, those in which customers will be able to gain access to complex information such as ordering, financial or directory information without human intervention. The capability for customers to identify themselves by giving name and address details and potentially to be verified using speaker recognition technology will allow progressively more complex applications to be automated, allowing agents to concentrate on higher value activities. This is discussed in detail elsewhere [4, 8].

7. Conclusions

The paper has discussed the integration of voice technology into the rapidly developing call centre market, focusing on the development of automatic customer interface (ACE) services for BT's residential customers.

Interactive voice response (IVR) is a key component in a call centre, being used for both automating call routing and transaction processing. The ACE services, based on the BT interactive speech platform, are addressing the latter.

Development and analysis of two typical services were discussed, highlighting the different dialogue strategies needed for general public applications and also closed user groups.

In future, customer contacts and transactions involving human agents and IVR systems will be complemented by the growing variety of on-line and interactive products such as Internet and interactive TV applications.

BT is strongly placed to exploit the technology within its own business applications, and to work with its business customers to design and deploy effective call centre and advanced voice solutions.

Acknowledgment

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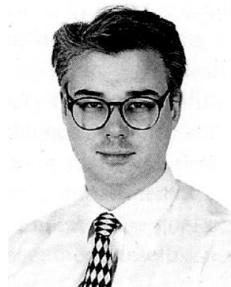
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Agostino Forcella joined BT in 1989, and was sponsored to follow a degree course. Having graduated with a first class honours degree in electrical and electronic engineering from Southbank University in 1992, he started at BT Laboratories, working on automated speech systems.

He has led the development of ACE trials since 1994.



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She led usability assessment tasks during the design and development of BT's CallMinder service.



Martin Hall joined BT Laboratories in 1986 after obtaining a BEng and PhD in electronic engineering from the University of Liverpool.

He worked on modem and speech processing projects, becoming head of the speech synthesis and analysis group in 1989 where he was responsible for producing the 'Laureate' text-to-speech system.

Since 1992, he has been project managing automated speech system development, and is currently a technical group leader with responsibility for delivering ACE systems and services.